

Lower Bounds on the Convergence Rate of Accept-Reject-Based Markov Chains

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Accept-Reject-Based Markov Chains

Any Markov kernel of the form

$$P(x, B) = \int_B a(x, y)q(y | x)d\lambda(y) + \delta_x(B)[1 - A(x)]$$

This includes, among others,

- Metropolis-Hastings

- Non-reversible Metropolis-Hastings

- Barker's

- Portkey Barker's

- Lazy Metropolis-Hastings

But the focus will be on Metropolis-Hastings.

Metropolis-Hastings

Let f be a target density and q_h a proposal density.

Given $X_t = x$, draw $Y \sim q_h(\cdot | x)$

Draw $U \sim \text{Unif}(0, 1)$ and set $X_{t+1} = y$ if

$$u \leq \frac{f(y)q_h(x | y)}{f(x)q_h(y | x)}$$

otherwise set $X_{t+1} = x$

Choice of h is crucial to finite-sample performance

Illustrative Example

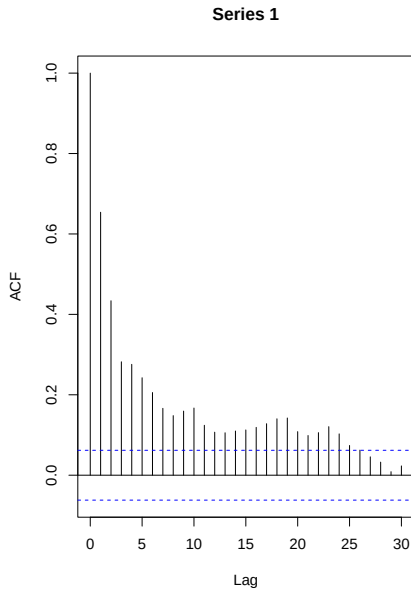
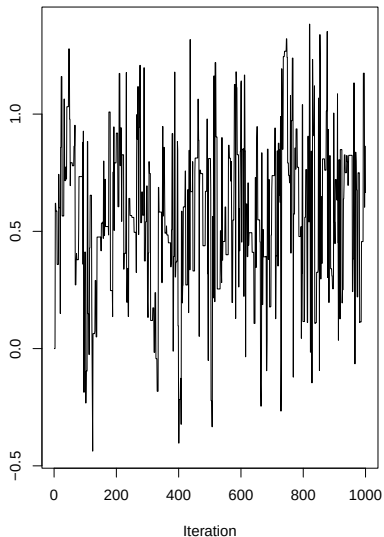
$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1), \quad \theta \sim N(0, 1)$$

$$\theta \mid x_1, \dots, x_n \sim N\left(\frac{n\bar{x}_n}{n+1}, \frac{1}{n+1}\right)$$

Use random walk Metropolis-Hastings with Normal proposal having variance h

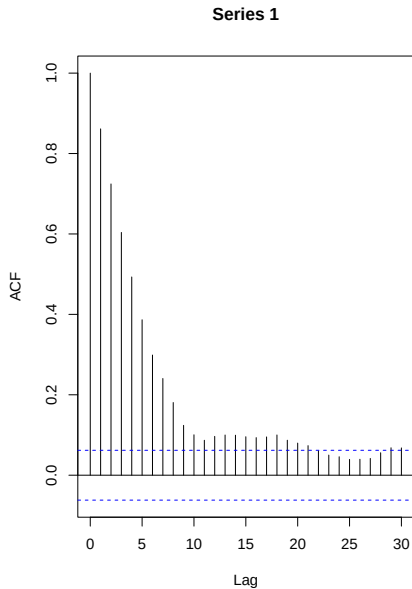
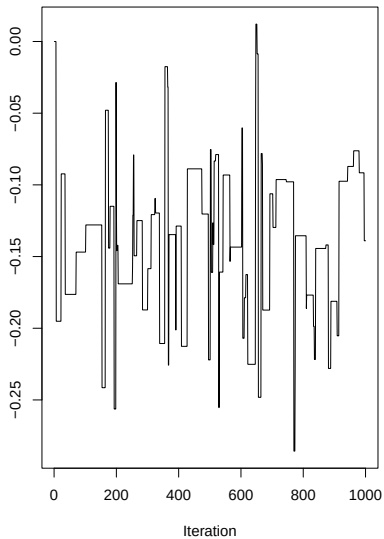
$n=2$, $h=1$

[1] 0.365



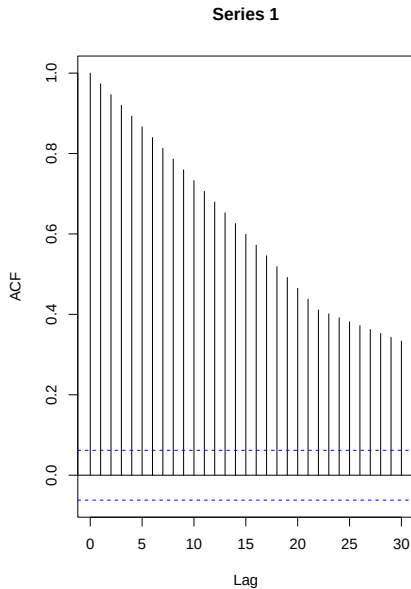
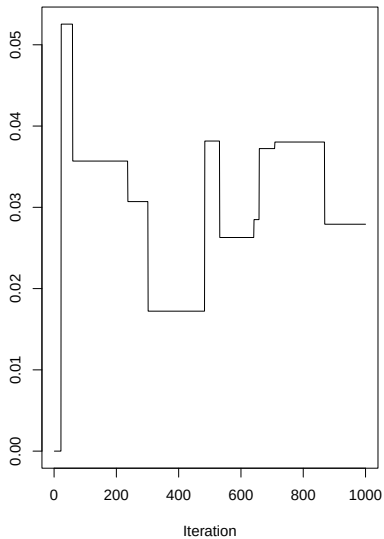
$n=20$, $h=1$

[1] 0.08



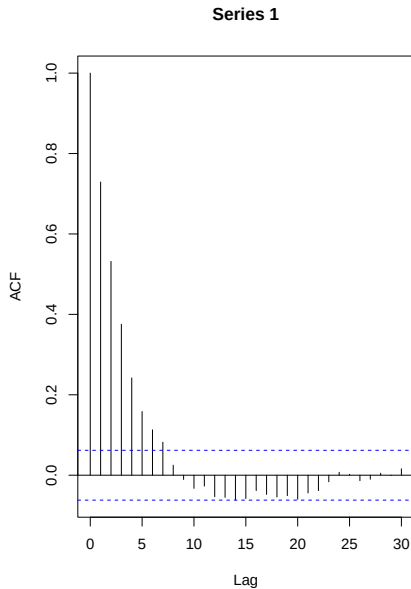
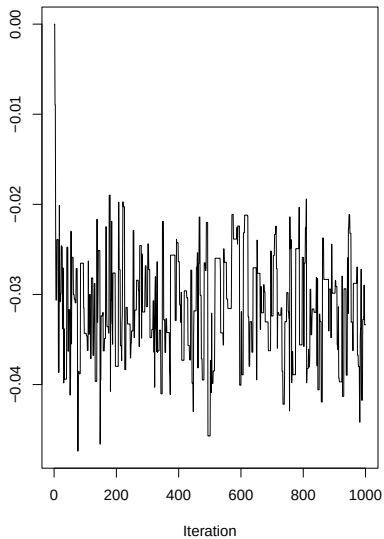
$n=200$, $h=1$

[1] 0.01



$n=200$, $h=1/2000$

[1] 0.281



Observations

The proposal scaling needs to account for the sample size and the dimension.

Significant theoretical efforts through optimal scaling arguments and mixing time analyses.

Practitioners largely use trial and error, basing their decisions on plots.

Can we identify when failure is guaranteed in general MH simulations so that we can avoid it?

Notation

The MH Markov kernel describes the dynamics of the Markov chain.
Informally,

$$\Pr(X_{t+j} \in B \mid X_j = x) = P^t(x, B)$$

$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \rightarrow 0 \quad t \rightarrow \infty$$

Question: When will this convergence take prohibitively long?

Lower Bound

Theorem If

$$A_h(x) = \int \left[\frac{f(x')q_h(x | x')}{f(x)q_h(x' | x)} \wedge 1 \right] q_h(x' | x) dx'.$$

then, for every x ,

$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \geq [1 - A_h(x)]^t$$

Answer: Avoid $A_h(x) \approx 0$.

Gaussian Proposals

Suppose $\mu : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and consider a proposal of the form

$$N_d(\mu(x), hC)$$

For example:

- Independence Sampler: $\mu(x) = c$
- Random Walk: $\mu(x) = x$
- MALA: $\mu(x) = x + h(\nabla \log x)/2$

Gaussian Proposals

Suppose $\mu : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and consider a proposal of the form

$$N_d(\mu(x), hC),$$

then

$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \geq \left[1 - \frac{1}{f(x)(2\pi h)^{d/2} \det(C)^{1/2}}\right]^t$$

Suggests

h must be small to avoid poor convergence properties

MH chains can have poor dimension dependence unless that scaling is chosen carefully:

$$h \propto d^{-\delta} \text{ for some } \delta > 0$$

Geometric Ergodicity

P is geometrically ergodic if there exists $\rho < 1$ such that

$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \leq M(x)\rho^t$$

Key sufficient condition for *reliable* MCMC

- Markov chain Central Limit Theorem

$$\sqrt{m}(\bar{X}_m - E_F(X)) \rightarrow N_p(0, \Sigma)$$

- Consistency of estimators of Σ such as batch means

Geometric Ergodicity

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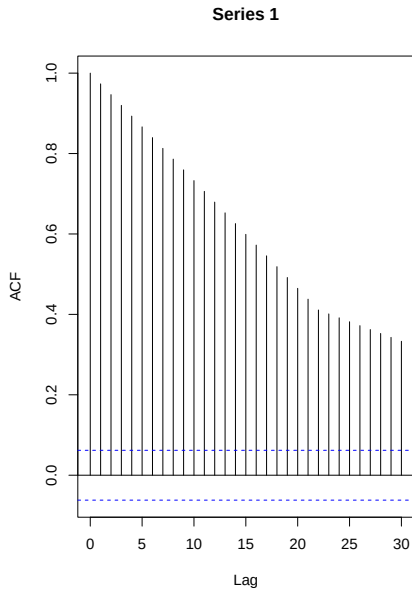
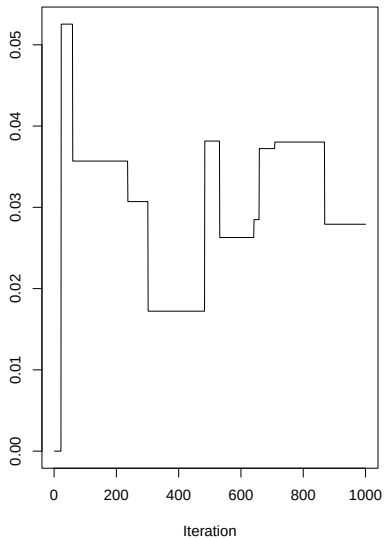
$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \leq M(x)\rho^t$$

Few results on constraining ρ for MH algorithms. That is, there are almost no results that identify constants such that

$$C_1 \leq \rho \leq C_2$$

Most results on geometric ergodicity of MH establish only the *existence* of $0 < \rho < 1$.

Illustrative Example $n=200$, $h=1$



Geometric Ergodicity

P is geometrically ergodic if there exists $\rho < 1$ such that

$$\|P^t(x, \cdot) - F(\cdot)\|_{TV} \leq M(x)\rho^t$$

Theorem If P is geometrically ergodic, then

$$\rho \geq 1 - \inf_x A_h(x)$$

Another Toy Example

Let $b > 1$ and

$$g(x, y) = \frac{b}{\pi} e^{-(x^2 + b^2 y^2)} \quad \text{and} \quad h(x, y) = \frac{b}{\pi} e^{-(b^2 x^2 + y^2)}.$$

Set

$$f(x, y) = \frac{1}{2}g(x, y) + \frac{1}{2}h(x, y)$$

RWMH using a Gaussian proposal with scaling h is geometrically ergodic.

Our result says

$$\rho \geq 1 - \frac{1}{2bh}$$

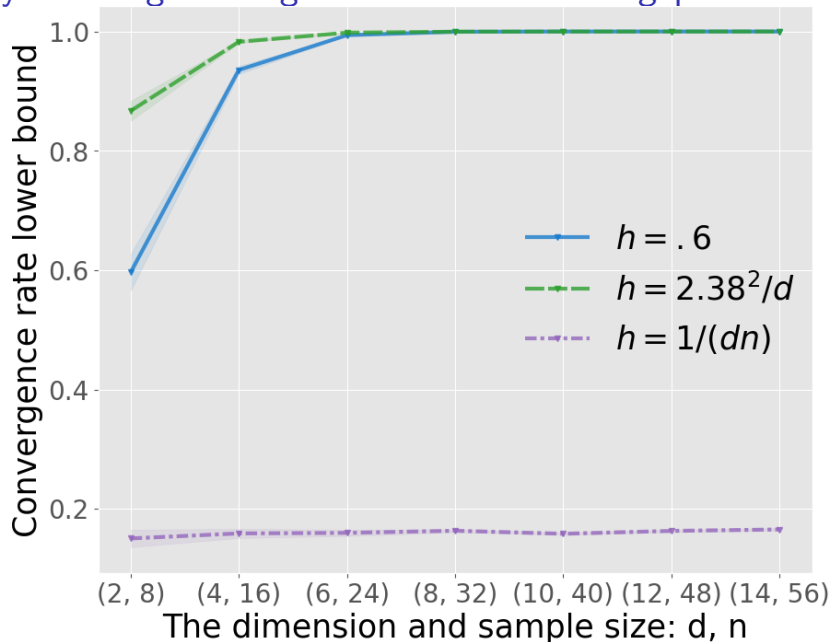
Bayesian Logistic Regression with Zellner's g -prior

$$f_n(\beta) \propto \prod_{i=1}^n s\left(\beta^T X_i\right)^{Y_i} \left(1 - s\left(\beta^T X_i\right)\right)^{1-Y_i} \exp\left(-\frac{1}{2g}\beta^T X^T X \beta\right).$$

Proposition Let β_n^* denote the point which maximizes π_n . If $n \rightarrow \infty$ in such a way that $d_n/n \rightarrow \gamma \in (0, 1)$, then, with probability 1, for all sufficiently large n , the acceptance probability for RWMH satisfies

$$\|P^t(\beta_n^*, \cdot) - F_n(\cdot)\|_{TV} \geq \left[1 - \left(\frac{hn(1 - \sqrt{\gamma})^2}{2g} + 1\right)^{-d_n/2}\right]^t.$$

Bayesian Logistic Regression with Zellner's g -prior



Bayesian Logistic Regression with a flat prior

Suppose for $i = 1, \dots, n$, (Y_i, X_i) are iid and

$$Y_i \mid X_i, \beta \stackrel{\text{ind}}{\sim} \text{Bern} \left(\left(1 + \exp \left(-\beta^T X_i \right) \right)^{-1} \right)$$

and $\nu(d\beta) = d\beta$.

Theorem If β_n^* maximizes the posterior and $d_n \leq n^\kappa$, $\kappa \in (0, 1)$, the posterior concentrates at β_n^* , under regularity conditions.

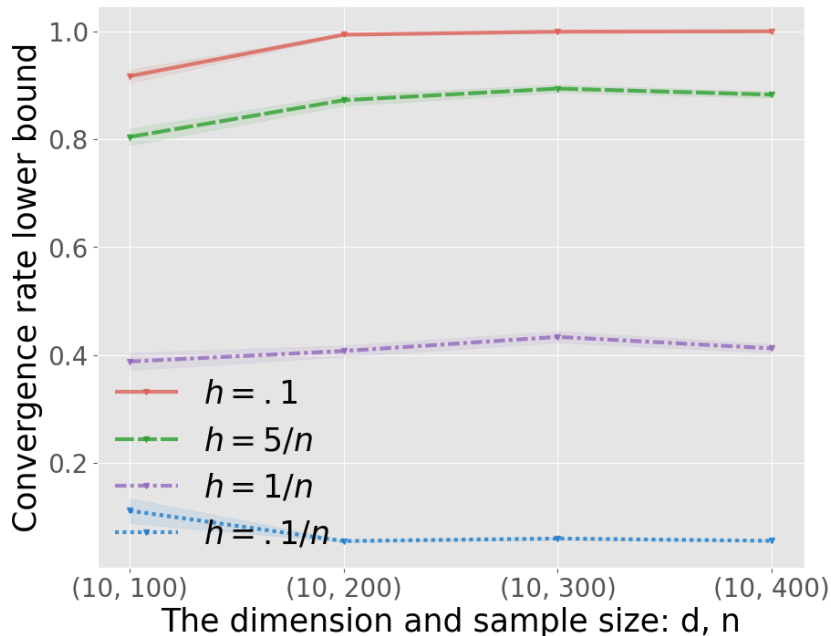
Bayesian Logistic Regression with a flat prior

If the proposal is $N_d(\mu(\beta), hC)$, then MH satisfies

$$\|P^t(\beta_n^*, \cdot) - F_n(\cdot)\|_{TV} \geq \left[1 - K \left(\frac{1}{nh}\right)^{d_n/2}\right]^t$$

Take Home Message: Proposal scaling needs to depend carefully on both n and d .

Bayesian Logistic Regression with a flat prior



What's in the paper

Lower bounds in both total variation and Wasserstein distances

Comparison to conductance methods

General lower bounds under posterior concentration

Existence of a spectral gap is equivalent to a geometric rate of convergence in many Wasserstein distances

Brown and Jones (2025) Lower Bounds on the Rate of Convergence for Accept-Reject-Based Markov Chains in Wasserstein and Total Variation Distances, *Bernoulli* DOI: 10.3150/24-BEJ1791